CELLS FOR COXETER GROUPS OF FINITE TYPE

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Contents

1. Introduction

The purpose of this note is to summarize the structure of the Kazhdan–Lusztig cells of Hecke algebras for finite Coxeter groups (corresponding to connected Coxeter graphs). For background on Kazhdan–Lusztig cells of Hecke algebras we refer to [Lus03].

Remark 1.1. We work in the case of equal parameters where Lusztig's conjectures P1–P15 in [Lus03, Conjectures 14.2] hold by e.g. categorification [EW14]. ▲

2. Conventions

Cells via matrices. Boxes correspond to (two-sided) cells, columns are left cells and rows are right cells. We write $\boldsymbol{k}_{n,m}$ for a $\mathbb{N}_0$-matrix with n rows and m columns containing only the entry k, e.g.

$$\mathbf{2}_{2,3} = \begin{array}{|c|c|c|}\hline 2 & 2 & 2 \\ \hline 2 & 2 & 2 \\ \hline \end{array} \quad \& \quad \mathbf{3}_{3,2} = \begin{array}{|c|c|}\hline 3 & 3 \\ \hline 3 & 3 \\ \hline 3 & 3 \\ \hline \end{array} \quad \& \quad \mathbf{1}_{3,3} = \begin{array}{|c|c|c|}\hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array} \tag{2.1}$$

The rightmost matrix in (2.1) in our notation corresponds to a (two-sided) cell containing 9 elements such that each intersection of a left and a right cell has precisely one element. Such a cell is called strongly regular and it is the easiest possible form a cell could have. Finally, note that for a strongly regular cell with matrix $\mathbf{1}_{n,n}$ one has that n is the square root of the size of the corresponding cell.

Another example for our conventions is

$$\begin{array}{|c|c|}\hline \mathbf{2}_{1,1} & \mathbf{1}_{1,2} \\ \hline \mathbf{1}_{2,1} & \mathbf{2}_{2,2} \\ \hline \end{array} = \begin{array}{|c|c|c|}\hline 2 & 1 & 1 \\ \hline 1 & 2 & 2 \\ \hline 1 & 2 & 2 \\ \hline \end{array} \rightsquigarrow \begin{cases} \text{three left and right cells,} \\ \text{intersections of sizes } 1 \text{ or } 2. \end{cases}$$

Longest elements. We color the boxes which contain longest elements of a parabolic subgroup, e.g.

$$\begin{array}{|c|c|}\hline \mathbf{2}_{1,1} & \mathbf{1}_{1,2} \\ \hline \mathbf{1}_{2,1} & \mathbf{2}_{2,2} \\ \hline \end{array} \tag{2.2}$$

means that the bottom right diagonal contains such elements, while the top left does not.

The asymptotic Hecke algebra. Fix a cell J. Lusztig's asymptotic Hecke algebra A_J (which he calls J-ring) appears in [Lus03, §18]. By Remark 1.1 it follows that for each left cell L (if no confusion can arise, then these are to be understood as being in J) there exists a Duflo involution which plays the role of the identity in the subalgebra $\mathrm{A}_{L\cap L^{-1}}$, where we call $L\cap L^{-1}$ the diagonal cell.

Choosing one diagonal cell, we thus write down the corresponding asymptotic Hecke algebra, e.g. in the example in (2.2)

$$\boxed{\mathbf{2}_{2,2}} \rightsquigarrow \mathrm{A} \tag{2.3}$$

means that the asymptotic Hecke algebra $\mathrm{A}_{L\cap L^{-1}}$ for this diagonal is isomorphic to the ring A.

Remark 2.1. For all left cells L_1, L_2 in each block one has

$$\mathrm{A}_{L_1\cap L_1^{-1}} \cong \mathrm{A}_{L_2\cap L_2^{-1}}$$

as rings, so the notation in (2.3) is well-defined. ▲

Clearly, for strongly regular cells J we have always $\mathrm{A}_{L\cap L^{-1}} \cong K_0(\mathcal{V}\mathrm{ec})\mathbb{Z}$ since $\#(L\cap L^{-1}) = 1$, so we omit those cases.

The asymptotic Hecke category. For a fixed cell J, there is also a notion of an asymptotic Hecke category $\mathcal{A}_J$ as it appears in [Lus03, §18, 18.15–18.20]. This category $\mathcal{A}_J$ decategorifies to A_J, and it also has a subcategory $\mathcal{A}_{L\cap L^{-1}}$ whose split Grothendieck group is $\mathrm{A}_{L\cap L^{-1}}$.

Remark 2.2. The categories $\mathcal{A}_J, \mathcal{A}_{L\cap L^{-1}}$ are defined in [Lus03, §18, 18.15–18.20] in terms of the Soergel bimodule framework from [EW14]. In general, $\mathcal{A}_J, \mathcal{A}_{L\cap L^{-1}}$ are known to be semisimple, monoidal categories. (This is a consequence of the construction in [Lus03, 18.15] and Soergel's hom formula [EW14, Theorem 3.6].) But we stress that, for Weyl groups, there is an alternative description using geometry which appears in several references which we use, e.g. [BFO09]. That these two definitions give equivalent monoidal categories is a consequence of [Soe90, Erweiterungssatz 5, Theorem 15]. We will use these two viewpoint interchangeably for Weyl groups. In fact, for Weyl groups, one can use the connection to geometry to prove that $\mathcal{A}_J, \mathcal{A}_{L\cap L^{-1}}$ are actually multi-fusion categories in the sense of [EGNO15, Definition 4.1.1], see [BFO09, Proof of Theorem 2] for an argument. ▲

In all cases, except one cell in type H_4, we identify the asymptotic Hecke algebra as a Grothendieck group of some category $\mathcal{A}$. If not stated otherwise, then the asymptotic Hecke category in this case is just $\mathcal{A}$. For Weyl groups this category is always a multi-fusion category by Remark 2.2, and, in general, its rank is just the rank of the associated asymptotic Hecke algebra. For example, for strongly regular cells we have always $\mathcal{A}_{L\cap L^{-1}} \cong \mathcal{V}\mathrm{ec}$.

Finally, for Weyl groups it actually follows from Remark 2.2 and e.g.[BFO09] that the stated asymptotic Hecke categories are the actual ones, while this is not a priori clear for dihedral types (see however, Remark 4.1) or types $\mathsf{H}_3, \mathsf{H}_4$.

Some notation for rings and categories. Here are some more rings and categories which we will need:

Definition 2.3. For a finite group G let $G\text{-}\mathcal{M}\mathrm{od}$ denote its fusion category of complex, finite-dimensional representations, and let $\mathcal{V}\mathrm{ec}_G$ the fusion category of G-graded vector spaces. (For both categories see e.g. [EGNO15, Example 4.1.2].) ▲

The Grothendieck rings $K_0(G\text{-}\mathcal{M}\mathrm{od})$ and $K_0(\mathcal{V}\mathrm{ec}_G) \cong \mathbb{Z}[G]$ will appear repeatedly below.

Definition 2.4. For $n \geq 2$ let $\mathrm{SU}(2)_n = \mathrm{U}_q\mathfrak{sl}_2\text{-}\mathcal{M}\mathrm{od}_n$ be the category of complex, finite-dimensional representations of quantum $\mathfrak{sl}_2$ semisimplified at level n. Let $\mathrm{SO}(3)_n$ denote the subcategory $\mathrm{U}_q\mathfrak{so}_3\text{-}\mathcal{M}\mathrm{od}_n$ which is the category of complex, finite-dimensional representations of quantum $\mathfrak{so}_3$ semisimplified at level n. (See e.g. [EGNO15, Example 8.18.5] for a discussion about those categories, which are actually modular.) ▲

Both categories from Definition 2.4 have well-studied Grothendieck rings $K_0(\mathrm{SU}(2)_n)$ and $K_0(\mathrm{SO}(3)_n)$ which are known as Verlinde rings.

Rank vectors. For a fixed multi-fusion category $\mathcal{C}$ we let $\mathrm{d}(\mathcal{C}) = (r_1, \dots, r_k)$ denote its vector of ranks of equivalence classes of simple transitive 2-representations.

Example 2.5. We have the following rank vectors for $S_i\text{-}\mathcal{M}\mathrm{od}$ in case $i = 3, 4, 5$.

$$\mathrm{d}(S_3\text{-}\mathcal{M}\mathrm{od}) = (1, 2, 3, 3),$$
$$\mathrm{d}(S_4\text{-}\mathcal{M}\mathrm{od}) = (1, 1, 1, 2, 2, 2, 3, 3, 3, 3, 4, 4, 4, 4, 5, 5),$$
$$\mathrm{d}(S_5\text{-}\mathcal{M}\mathrm{od}) = (1, 1, 1, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5, 6, 6, 7),$$

which are known, e.g. it is a consequence of [EGNO15, Corollary 7.12.20]. ▲

Example 2.6. In case of the Verlinde categories we have

$$\mathrm{d}(\mathrm{SO}(3)_n) = (\tfrac{n-1}{2}), \quad \text{for } n \text{ odd},$$
$$\mathrm{d}(\mathrm{SO}(3)_n) = (\lfloor \tfrac{n+2}{4} \rfloor - 1, \lceil \tfrac{n+2}{4} \rceil + 1, \tfrac{n-2}{2}, \tfrac{n}{2}), \quad \text{for } n \text{ odd}, n \neq 4, 12, 18, 30,$$

$$\mathrm{d}(\mathrm{SO}(3)_4) = (1,2) \quad \& \quad \mathrm{d}(\mathrm{SO}(3)_{12}) = (3,3,3,4,5,6)$$
$$\mathrm{d}(\mathrm{SO}(3)_{18}) = (3,4,4,6,8,9) \quad \& \quad \mathrm{d}(\mathrm{SO}(3)_{30}) = (4,4,7,9,14,15),$$

which follows from the usual ADE classification, cf. [KO02]. ▲

To compute these vectors, the following is useful:

Definition 2.7. For a finite group G let $H^2(G,\mathbb{C}^*)$ be its group of Schur multipliers. ▲

These Schur multipliers play an important role: they index simple transitive 2-representations (up to equivalence) of $\mathcal{V}\mathrm{ec}_G$ since these are in bijection with

$$\{(H,\phi) \mid H \subset G \text{ subgroup}, \phi \in H^2(G,\mathbb{C}^*)\}.$$

Under this bijection the rank of simple transitive 2-representations corresponds to the relative sizes $|G/H|$ of the subgroups H, cf. [EGNO15, Example 7.4.10].

Example 2.8. Because $H^2(\mathbb{Z}/2\mathbb{Z},\mathbb{C}^*) \cong 1$ we see that $\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}$ has two simple transitive 2-representations (up to equivalence), and $\mathrm{d}(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}) = (1,2)$. ▲

Example 2.9. More general than Example 2.8, we have $H^2((\mathbb{Z}/2\mathbb{Z})^d,\mathbb{C}^*) \cong (\mathbb{Z}/2\mathbb{Z})^s$ for $d \in \mathbb{Z}_{>0}$ and $s = \frac{d(d-1)}{2}$. In particular, $\mathrm{d}(\mathcal{V}\mathrm{ec}_{(\mathbb{Z}/2\mathbb{Z})^2}) = (1,1,2,2,2,4)$. ▲

Example 2.10. Since $H^2(S_3,\mathbb{C}^*) \cong 1$, we get $\mathrm{d}(\mathcal{V}\mathrm{ec}_{S_3}) = (1,2,3,6)$ which looks very similar to $\mathrm{d}(S_3\text{-}\mathcal{M}\mathrm{od}) = (1,2,3,3)$ from Example 2.5. In fact, for any finite group, $\mathcal{V}\mathrm{ec}_G$ and G-$\mathcal{M}\mathrm{od}$ are categorically Morita equivalent in the sense of [EGNO15, Definition 7.12.17]. ▲

3. Rank one

There is only one Coxeter group of rank one.

3.1. Type A_1. Graph:

$$1$$

Number of elements: 2. Number of cells: 2, named 0 (trivial) to 1 (top). Number of parabolic subgroups: 2.

Cell order:

$$0$$

Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1
size	1	1
a	0	1
p	1	1
sr	yes	yes

4. Rank two

Recall that rank two Coxeter groups are all dihedral groups for $n \geq 2$.

4.1. **Type** I_{2n}**.** Graph:

$$1 \overset{n}{\textbf{—}} 2$$

Number of elements: $2n$. Number of cells: 3, named $\mathtt{0}$ (trivial) to $\mathtt{2}$ (top). Number of parabolic subgroups: 4.

Cell order:

$$\mathtt{0} - \mathtt{1} - \mathtt{2}$$

Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2
size	1	$2n{-}2$	1
a	0	1	n
p	1	2	1
sr	yes	$*$	yes

Here $*$ means that the cell is strongly regular if and only if $n = 3$.

Cell $\mathtt{1}$ for n being even or odd is

even: $\begin{array}{|c|c|}\hline \frac{n}{2} & \frac{n-2}{2} \\ \hline \frac{n-2}{2} & \frac{n}{2} \\ \hline \end{array}$ $\boxed{\frac{n}{2}}$ $\rightsquigarrow K_0(\mathrm{SO}(3)_n)$, odd: $\boldsymbol{\frac{n-1}{2}}_{2,2} = \begin{array}{|c|c|}\hline \frac{n-1}{2} & \frac{n-1}{2} \\ \hline \frac{n-1}{2} & \frac{n-1}{2} \\ \hline \end{array}$ $\boxed{\frac{n-1}{2}}$ $\rightsquigarrow K_0(\mathrm{SO}(3)_n)$.

Remark 4.1. Although the dihedral groups are in general not Weyl groups, it is easy to see that the asymptotic Hecke category is actually the multi-fusion category $\mathrm{SO}(3)_n$. ▲

The rank vectors of simple transitive 2-representations associated to the asymptotic Hecke categories $\mathrm{SO}(3)_n$ in these cases are given in Example 2.6.

5. Rank three

There are three finite Coxeter groups of rank three.

5.1. **Type** A_3**.** Graph:

$$1 - 2 - 3$$

Number of elements: 24. Number of cells: 5, named $\mathtt{0}$ (trivial) to $\mathtt{4}$ (top). Number of parabolic subgroups: 8.

Cell order:

$$\mathtt{0} - \mathtt{1} - \mathtt{2} - \mathtt{3} - \mathtt{4}$$

Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4
size	1	9	4	9	1
a	0	1	2	3	6
p	1	3	2	1	1
sr	yes	yes	yes	yes	yes

5.2. **Type** B_3**.** Graph:

$$1 \overset{4}{\text{—}} 2 \text{—} 3$$

Number of elements: 48. Number of cells: 6, named 0 (trivial) to 5 (top). Number of parabolic subgroups: 8.

Cell order:

$$0 \text{—} 1 \text{—} 2 \text{—} 3 \text{—} 4 \text{—} 5$$

Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5
size	1	14	9	9	14	1
a	0	1	2	3	4	9
p	1	3	1	1	1	1
sr	yes	no	yes	yes	no	yes

Cells 1 and 4 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{1,1} & \mathbf{1}_{1,2} \\ \hline \mathbf{1}_{2,1} & \mathbf{2}_{2,2} \\ \hline \end{array} \qquad \mathbf{2}_{1,1} \ \& \ \mathbf{2}_{2,2} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{1,1} & \mathbf{1}_{1,2} \\ \hline \mathbf{1}_{2,1} & \mathbf{2}_{2,2} \\ \hline \end{array} \qquad \mathbf{1}_{2,2} \ \& \ \mathbf{2}_{2,2} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

The rank vectors in these case can be obtained from Example 2.8.

5.3. **Type** H_3**.** Graph:

$$1 \overset{5}{\text{—}} 2 \text{—} 3$$

Number of elements: 120. Number of cells: 7, named 0 (trivial) to 6 (top). Number of parabolic subgroups: 8.

Cell order:

$$0 \text{—} 1 \text{—} 2 \text{—} 3 \text{—} 4 \text{—} 5 \text{—} 6$$

Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6
size	1	18	25	32	25	18	1
a	0	1	2	3	5	6	15
p	1	3	1	1	1	0	1
sr	yes	no	yes	no	yes	no	yes

Cells 1 and 5 are:

$$\mathbf{2}_{3,3} \quad \mathbf{2}_{3,3} \rightsquigarrow K_0(\mathrm{SO}(3)_5), \qquad \mathbf{2}_{3,3} \quad \mathbf{2}_{3,3} \rightsquigarrow K_0(\mathrm{SO}(3)_5).$$

Cell 3 is:

$$\mathbf{2}_{4,4} \quad \mathbf{2}_{4,4} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Remark 5.1. Here it is not clear that the asymptotic Hecke category is $\mathrm{SO}(3)_5$ or $\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}$. For instance, $K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}})$ has two possible categorifications. ▲

Remark 5.2. This is the first example outside of dihedral type of a discrepancy of the number of simple transitive 2-representations of $\mathrm{SO}(3)_5$ – which has two of rank 2 – and the ones one can read of from $\mathbf{2}_{3,3}$. Similarly, $\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}$ has two simple transitive 2-representations, one of rank 1 and one of rank 2, but we have the cell matrix $\mathbf{2}_{4,4}$. ▲

6. Rank four

In rank four there are five finite Coxeter groups, which is the maximum among all ranks.

6.1. **Type** A_4. Graph:

$$1 — 2 — 3 — 4$$

Number of elements: 120. Number of cells: 7, named 0 (trivial) to 6 (top). Number of parabolic subgroups: 16.

Cell order:

$$0 — 1 — 2 — 3 — 4 — 5 — 6$$

Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

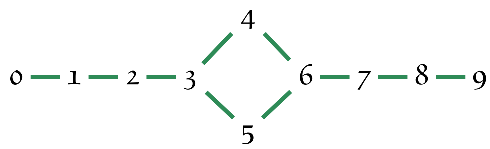
cell	0	1	2	3	4	5	6
size	1	16	25	36	25	16	1
a	0	1	2	3	4	6	10
p	1	4	3	3	2	2	1
sr	yes	yes	yes	yes	yes	yes	yes

6.2. **Type** B_4. Graph:

$$1 \overset{4}{—} 2 — 3 — 4$$

Number of elements: 384. Number of cells: 10, named 0 (trivial) to 9 (top). Number of parabolic subgroups: 16.

Cell order:



Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6	7	8	9
size	1	26	56	64	54	36	64	56	26	1
a	0	1	2	3	4	4	5	6	9	16
p	1	4	3	2	1	1	1	1	1	1
sr	yes	no	no	yes	no	yes	yes	no	no	yes

Cells 1 and 8 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{1,1} & \mathbf{1}_{1,3} \\ \hline \mathbf{1}_{3,1} & \mathbf{2}_{3,3} \\ \hline \end{array} \qquad \mathbf{2}_{1,1} \;\&\; \mathbf{2}_{3,3} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{1,1} & \mathbf{1}_{1,3} \\ \hline \mathbf{1}_{3,1} & \mathbf{2}_{3,3} \\ \hline \end{array} \qquad \mathbf{2}_{1,1} \;\&\; \mathbf{2}_{3,3} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 2 and 7 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{2,2} & \mathbf{1}_{2,4} \\ \hline \mathbf{1}_{4,2} & \mathbf{2}_{4,4} \\ \hline \end{array} \qquad \mathbf{2}_{2,2} \;\&\; \mathbf{2}_{4,4} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

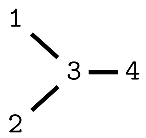
$$\begin{array}{|c|c|}\hline \mathbf{2}_{2,2} & \mathbf{1}_{2,4} \\ \hline \mathbf{1}_{4,2} & \mathbf{2}_{4,4} \\ \hline \end{array} \qquad \mathbf{2}_{2,2} \;\&\; \mathbf{2}_{4,4} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

Cell 4 is:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{3,3} & \mathbf{1}_{3,3}\\ \hline \mathbf{1}_{3,3} & \mathbf{2}_{3,3}\\ \hline\end{array} \qquad \mathbf{2}_{3,3}\ \&\ \mathbf{2}_{3,3} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

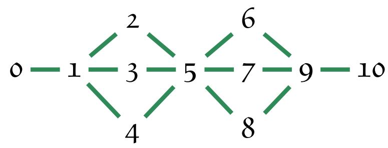
For the rank vectors see Example 2.8.

6.3. **Type** D_4. Graph:



Number of elements: 192. Number of cells: 11, named 0 (trivial) to 10 (top). Number of parabolic subgroups: 16.

Cell order:



Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6	7	8	9	10
size	1	16	9	9	9	104	9	9	9	16	1
a	0	1	2	2	2	3	6	6	6	7	12
p	1	4	1	1	1	4	1	1	1	0	1
sr	yes	yes	yes	yes	yes	no	yes	yes	yes	yes	yes

Cell 5 is:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{2,2} & \mathbf{1}_{2,6}\\ \hline \mathbf{1}_{6,2} & \mathbf{2}_{6,6}\\ \hline\end{array} \qquad \mathbf{2}_{2,2}\ \&\ \mathbf{2}_{6,6} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

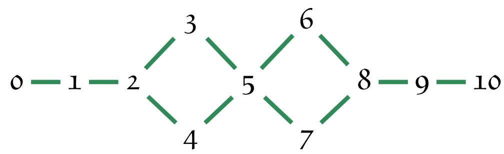
For the rank vectors see Example 2.8.

6.4. **Type** F_4. Graph:

$$1 - 2 = 3 - 4$$

Number of elements: 1152. Number of cells: 11, named 0 (trivial) to 10 (top). Number of parabolic subgroups: 16.

Cell order:



Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6	7	8	9	10
size	1	24	81	64	64	684	64	64	81	24	1
a	0	1	2	3	3	4	9	9	10	13	24
p	1	4	3	1	1	3	1	1	0	0	1
sr	yes	no	yes	yes	yes	no	yes	yes	yes	no	yes

Cells 1 and 9 are:

$\mathbf{2}_{2,2}$	$\mathbf{1}_{2,2}$
$\mathbf{1}_{2,2}$	$\mathbf{2}_{2,2}$

$\mathbf{2}_{2,2} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}})$,

$\mathbf{2}_{2,2}$	$\mathbf{1}_{2,2}$
$\mathbf{1}_{2,2}$	$\mathbf{2}_{2,2}$

$\mathbf{2}_{2,2} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}})$.

Cell 5 is:

$\mathbf{5}_{3,3}$	$\mathbf{3}_{3,3}$	$\mathbf{4}_{3,4}$	$\mathbf{5}_{3,1}$	$\mathbf{2}_{3,1}$
$\mathbf{3}_{3,3}$	$\mathbf{5}_{3,3}$	$\mathbf{4}_{3,4}$	$\mathbf{2}_{3,1}$	$\mathbf{5}_{3,1}$
$\mathbf{4}_{4,3}$	$\mathbf{4}_{4,3}$	$\mathbf{9}_{4,4}$	$\mathbf{6}_{4,1}$	$\mathbf{6}_{4,1}$
$\mathbf{5}_{1,3}$	$\mathbf{2}_{1,3}$	$\mathbf{6}_{1,4}$	$\mathbf{9}_{1,1}$	$\mathbf{3}_{1,1}$
$\mathbf{2}_{1,3}$	$\mathbf{5}_{1,3}$	$\mathbf{6}_{1,4}$	$\mathbf{3}_{1,1}$	$\mathbf{9}_{1,1}$

$\mathbf{5}_{3,3} \leftrightsquigarrow K_0(S_4\text{-}\mathcal{M}\mathrm{od})$,

$\mathbf{9}_{4,4} \leftrightsquigarrow K_0(G\text{-}\mathcal{M}\mathrm{od})$,

$\mathbf{9}_{1,1} \leftrightsquigarrow K_0(G'\text{-}\mathcal{M}\mathrm{od})$,

where $G = S_4/(S_2{\times}S_2){\times}S_4/(S_2{\times}S_2)$ and $G' = S_4/S_2{\times}S_4/S_2$.

Except for the cell 5, the rank vectors are in Example 2.8. For 5 consider Example 2.5.

Remark 6.1. Note that this is the first case where we see two non-isomorphic asymptotic Hecke algebras for diagonals of the same cell. One of them, namely $K_0(G\text{-}\mathcal{M}\mathrm{od})$, is not commutative, and can thus not be categorified by a modular category. Similarly, according to the classification of modular categories of rank 5, the multi-fusion category $S_4\text{-}\mathcal{M}\mathrm{od}$ can not be enriched into a modular category. ▲

6.5. **Type** H_4**.** Graph:

$$1 \overset{5}{—} 2 — 3 — 4$$

Number of elements: 14400. Number of cells: 13, named 0 (trivial) to 12 (top). Number of parabolic subgroups: 16.

Cell order:

$$0 — 1 — 2 — 3 — 4 — 5 — 6 — 7 — 8 — 9 — 10 — 11 — 12$$

Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6	7	8	9	10	11	12
size	1	32	162	512	625	1296	9144	1296	625	512	162	32	1
a	0	1	2	3	4	5	6	15	16	18	22	31	60
p	1	4	3	2	1	1	2	1	0	0	0	0	1
sr	yes	no	no	no	yes	yes	no	yes	yes	no	no	no	yes

Cells 1 and 11 are:

$\mathbf{2}_{4,4}$ $\quad \mathbf{2}_{4,4} \leftrightsquigarrow K_0(\mathrm{SO}(3)_5)$, $\qquad \mathbf{2}_{4,4}$ $\quad \mathbf{2}_{4,4} \leftrightsquigarrow K_0(\mathrm{SO}(3)_5)$.

Cells 2 and 10 are:

$\mathbf{2}_{9,9}$ $\quad \mathbf{2}_{9,9} \leftrightsquigarrow K_0(\mathrm{SO}(3)_5)$, $\qquad \mathbf{2}_{9,9}$ $\quad \mathbf{2}_{9,9} \leftrightsquigarrow K_0(\mathrm{SO}(3)_5)$.

Cells 3 and 9 are:

$\mathbf{2}_{16,16}$ $\quad \mathbf{2}_{16,16} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}})$, $\qquad \mathbf{2}_{16,16}$ $\quad \mathbf{2}_{16,16} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}})$.

Remark 6.2. For these cells the same as in Remarks 5.1 and 5.2 holds. ▲

Cell 6 is:

$\mathbf{14}_{8,8}$	$\mathbf{13}_{10,8}$	$\mathbf{14}_{6,8}$
$\mathbf{13}_{8,10}$	$\mathbf{18}_{10,10}$	$\mathbf{18}_{6,10}$
$\mathbf{14}_{8,6}$	$\mathbf{18}_{10,6}$	$\mathbf{24}_{6,6}$

$$\mathbf{14}_{8,8} \rightsquigarrow \mathrm{A}_1,\quad \mathbf{18}_{10,10} \rightsquigarrow \mathrm{A}_2,\quad \mathbf{24}_{6,6} \rightsquigarrow \mathrm{A}_3$$

where the three rings $\mathrm{A}_1, \mathrm{A}_2$ and A_3 can be explicitly computed, but, to the best of our knowledge, do not have any nice presentation. Note that none of these is commutative, cf. [Alv08].

Remark 6.3. This is the only case where we are not sure about any potential categorification of the asymptotic Hecke algebras. ▲

7. Rank five

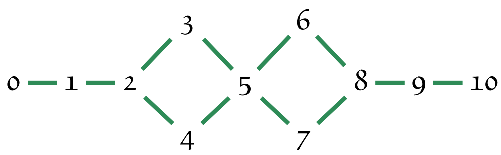
In rank five there are three finite Coxeter groups.

7.1. Type A_5. Graph:

$$1 - 2 - 3 - 4 - 5$$

Number of elements: 600. Number of cells: 11, named 0 (trivial) to 10 (top). Number of parabolic subgroups: 32.

Cell order:



Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

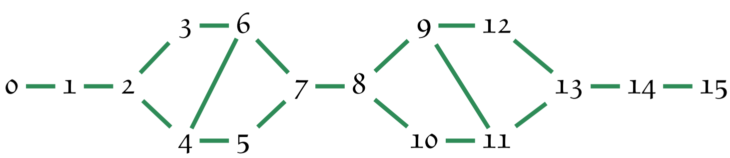
cell	0	1	2	3	4	5	6	7	8	9	10
size	1	25	81	100	25	256	100	25	81	25	1
a	0	1	2	3	3	4	6	6	7	10	15
p	1	5	6	4	1	6	3	1	2	2	1
sr	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes

7.2. Type B_5. Graph:

$$1 \overset{4}{-} 2 - 3 - 4 - 5$$

Number of elements: 3840. Number of cells: 16, named 0 (trivial) to 15 (top). Number of parabolic subgroups: 32.

Cell order:



Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
size	1	42	150	100	225	152	600	650	650	600	152	225	100	150	42	1
a	0	1	2	3	3	4	4	5	6	7	9	10	10	11	16	25
p	1	5	6	1	3	1	4	2	2	2	1	1	1	0	1	1
sr	yes	no	no	yes	yes	no	no	no	no	no	no	yes	yes	no	no	yes

Cells 1 and 14 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{1,1} & \mathbf{1}_{4,1} \\ \hline \mathbf{1}_{1,4} & \mathbf{2}_{4,4} \\ \hline \end{array} \quad \mathbf{2}_{1,1} \ \& \ \mathbf{2}_{4,4} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{1,1} & \mathbf{1}_{4,1} \\ \hline \mathbf{1}_{1,4} & \mathbf{2}_{4,4} \\ \hline \end{array} \quad \mathbf{2}_{1,1} \ \& \ \mathbf{2}_{4,4} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 2 and 13 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{5,5} & \mathbf{1}_{5,5} \\ \hline \mathbf{1}_{5,5} & \mathbf{2}_{5,5} \\ \hline \end{array} \quad \mathbf{2}_{5,5} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{5,5} & \mathbf{1}_{5,5} \\ \hline \mathbf{1}_{5,5} & \mathbf{2}_{5,5} \\ \hline \end{array} \quad \mathbf{2}_{5,5} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 5 and 10 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{4,4} & \mathbf{1}_{6,4} \\ \hline \mathbf{1}_{4,6} & \mathbf{2}_{6,6} \\ \hline \end{array} \quad \mathbf{2}_{4,4} \ \& \ \mathbf{2}_{6,6} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{4,4} & \mathbf{1}_{6,4} \\ \hline \mathbf{1}_{4,6} & \mathbf{2}_{6,6} \\ \hline \end{array} \quad \mathbf{2}_{4,4} \ \& \ \mathbf{2}_{6,6} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 6 and 9 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{10,10} & \mathbf{1}_{10,10} \\ \hline \mathbf{1}_{10,10} & \mathbf{2}_{10,10} \\ \hline \end{array} \quad \mathbf{2}_{10,10} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{10,10} & \mathbf{1}_{10,10} \\ \hline \mathbf{1}_{10,10} & \mathbf{2}_{10,10} \\ \hline \end{array} \quad \mathbf{2}_{10,10} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

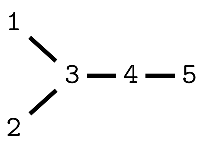
Cells 7 and 8 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{5,5} & \mathbf{1}_{15,5} \\ \hline \mathbf{1}_{5,15} & \mathbf{2}_{15,15} \\ \hline \end{array} \quad \mathbf{2}_{5,5} \ \& \ \mathbf{2}_{15,15} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{5,5} & \mathbf{1}_{15,5} \\ \hline \mathbf{1}_{5,15} & \mathbf{2}_{15,15} \\ \hline \end{array} \quad \mathbf{2}_{5,5} \ \& \ \mathbf{2}_{15,15} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

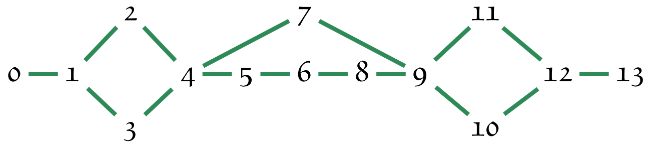
Again, the rank vectors can be found in Example 2.8.

7.3. **Type** D_5**.** Graph:



Number of elements: 1920. Number of cells: 14, named 0 (trivial) to 13 (top). Number of parabolic subgroups: 32.

Cell order:



Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6	7	8	9	10	11	12	13
size	1	25	16	100	350	400	100	36	400	350	100	16	25	1
a	0	1	2	2	3	4	5	6	6	7	10	12	13	20
p	1	5	1	5	6	4	1	1	3	1	2	1	0	1
sr	yes	yes	yes	yes	no	yes	yes	yes	yes	no	yes	yes	yes	yes

Cells 4 and 9 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{5,5} & \mathbf{1}_{10,5} \\ \hline \mathbf{1}_{5,10} & \mathbf{2}_{10,10} \\ \hline \end{array} \quad \mathbf{2}_{5,5} \,\&\, \mathbf{2}_{10,10} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{5,5} & \mathbf{1}_{10,5} \\ \hline \mathbf{1}_{5,10} & \mathbf{2}_{10,10} \\ \hline \end{array} \quad \mathbf{2}_{5,5} \,\&\, \mathbf{2}_{10,10} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

As before, the rank vectors can be found in Example 2.8.

8. Rank six

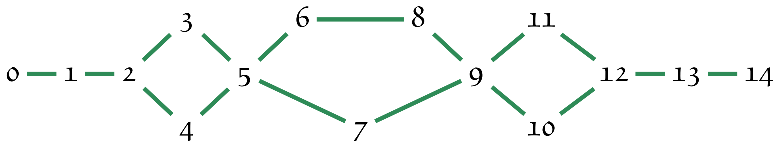
There are four finite Coxeter groups of rank six.

8.1. **Type** A_6. Graph:

$$1 - 2 - 3 - 4 - 5 - 6$$

Number of elements: 600. Number of cells: 15, named 0 (trivial) to 14 (top). Number of parabolic subgroups: 64.

Cell order:



Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

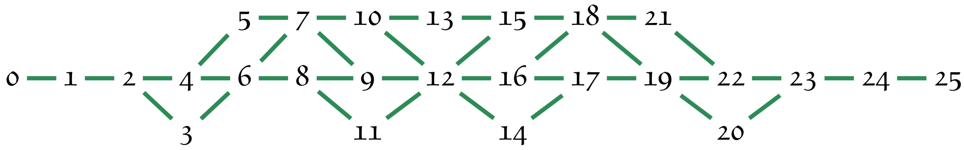
cell	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
size	1	36	196	225	196	1225	441	400	441	1225	196	225	196	36	1
a	0	1	2	3	3	4	5	6	6	7	9	10	11	15	21
p	1	6	10	5	4	12	3	4	3	6	2	3	2	2	1
sr	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes

8.2. **Type** B_6. Graph:

$$1 \overset{4}{-} 2 - 3 - 4 - 5 - 6$$

Number of elements: 46080. Number of cells: 26, named 0 (trivial) to 25 (top). Number of parabolic subgroups: 64.

Cell order:



Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
size	1	62	342	576	650	3150	350	1600	2432	3402	900	2025	14500	600	2025	900	3402	2432	1600	350	576	3150	650	342	62	1
a	0	1	2	3	3	4	4	5	5	6	6	6	7	9	10	10	10	15	11	16	17	12	15	25	25	36
p	1	6	10	4	4	9	1	2	3	3	1	1	6	1	2	1	2	0	1	1	1	1	1	0	1	1
sr	yes	no	no	yes	no	no	no	yes	no	no	yes	yes	no	no	yes	yes	no	no	yes	no	yes	no	no	no	no	yes

Cells 1 and 24 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{1,1} & \mathbf{1}_{1,5} \\ \hline \mathbf{1}_{5,1} & \mathbf{2}_{5,5} \\ \hline \end{array} \qquad \mathbf{2}_{1,1} \;\&\; \mathbf{2}_{5,5} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{1,1} & \mathbf{1}_{1,5} \\ \hline \mathbf{1}_{5,1} & \mathbf{2}_{5,5} \\ \hline \end{array} \qquad \mathbf{2}_{1,1} \;\&\; \mathbf{2}_{5,5} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 2 and 23 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{6,6} & \mathbf{1}_{6,9} \\ \hline \mathbf{1}_{9,6} & \mathbf{2}_{9,9} \\ \hline \end{array} \qquad \mathbf{2}_{6,6} \;\&\; \mathbf{2}_{9,9} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{6,6} & \mathbf{1}_{6,9} \\ \hline \mathbf{1}_{9,6} & \mathbf{2}_{9,9} \\ \hline \end{array} \qquad \mathbf{2}_{6,6} \;\&\; \mathbf{2}_{9,9} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 4 and 22 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{5,5} & \mathbf{1}_{5,15} \\ \hline \mathbf{1}_{15,5} & \mathbf{2}_{15,15} \\ \hline \end{array} \qquad \mathbf{2}_{5,5} \;\&\; \mathbf{2}_{15,15} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{5,5} & \mathbf{1}_{5,15} \\ \hline \mathbf{1}_{15,5} & \mathbf{2}_{15,15} \\ \hline \end{array} \qquad \mathbf{2}_{5,5} \;\&\; \mathbf{2}_{15,15} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 5 and 21 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{15,15} & \mathbf{1}_{15,30} \\ \hline \mathbf{1}_{30,15} & \mathbf{2}_{30,30} \\ \hline \end{array} \qquad \mathbf{2}_{15,15} \;\&\; \mathbf{2}_{30,30} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{15,15} & \mathbf{1}_{15,30} \\ \hline \mathbf{1}_{30,15} & \mathbf{2}_{30,30} \\ \hline \end{array} \qquad \mathbf{2}_{15,15} \;\&\; \mathbf{2}_{30,30} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 6 and 19 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{5,5} & \mathbf{1}_{5,10} \\ \hline \mathbf{1}_{10,5} & \mathbf{2}_{10,10} \\ \hline \end{array} \qquad \mathbf{2}_{5,5} \;\&\; \mathbf{2}_{10,10} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{5,5} & \mathbf{1}_{5,10} \\ \hline \mathbf{1}_{10,5} & \mathbf{2}_{10,10} \\ \hline \end{array} \qquad \mathbf{2}_{5,5} \;\&\; \mathbf{2}_{10,10} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 8 and 17 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{16,16} & \mathbf{1}_{16,24} \\ \hline \mathbf{1}_{24,16} & \mathbf{2}_{24,24} \\ \hline \end{array} \qquad \mathbf{2}_{16,16} \;\&\; \mathbf{2}_{24,24} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|}\hline \mathbf{2}_{16,16} & \mathbf{1}_{16,24} \\ \hline \mathbf{1}_{24,16} & \mathbf{2}_{24,24} \\ \hline \end{array} \qquad \mathbf{2}_{16,16} \;\&\; \mathbf{2}_{24,24} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 9 and 16 are:

$$\begin{array}{|c|c|}\hline \mathbf{2}_{9,9} & \mathbf{1}_{9,36} \\ \hline \mathbf{1}_{36,9} & \mathbf{2}_{36,36} \\ \hline \end{array} \qquad \mathbf{2}_{9,9} \;\&\; \mathbf{2}_{36,36} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$\mathbf{2}_{9,9}$	$\mathbf{1}_{9,36}$
$\mathbf{1}_{36,9}$	$\mathbf{2}_{36,36}$

$\mathbf{2}_{9,9}$ & $\mathbf{2}_{36,36}$ $\rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}})$.

Cell 12 is:

$\mathbf{4}_{5,5}$	$\mathbf{1}_{5,5}$	$\mathbf{1}_{5,20}$	$\mathbf{2}_{5,25}$	$\mathbf{2}_{5,25}$
$\mathbf{1}_{5,5}$	$\mathbf{4}_{5,5}$	$\mathbf{1}_{5,20}$	$\mathbf{2}_{5,25}$	$\mathbf{2}_{5,25}$
$\mathbf{1}_{20,5}$	$\mathbf{1}_{20,5}$	$\mathbf{4}_{20,20}$	$\mathbf{2}_{20,25}$	$\mathbf{2}_{20,25}$
$\mathbf{2}_{25,5}$	$\mathbf{2}_{25,5}$	$\mathbf{2}_{25,20}$	$\mathbf{4}_{25,25}$	$\mathbf{1}_{25,25}$
$\mathbf{2}_{25,5}$	$\mathbf{2}_{25,5}$	$\mathbf{2}_{25,20}$	$\mathbf{1}_{25,25}$	$\mathbf{4}_{25,25}$

$\mathbf{4}_{5,5}$ & $\mathbf{4}_{20,20}$ & $\mathbf{4}_{25,25}$ $\rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}})$.

Cell 13 is:

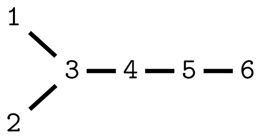
$\mathbf{2}_{10,10}$	$\mathbf{1}_{10,10}$
$\mathbf{1}_{10,10}$	$\mathbf{2}_{10,10}$

$\mathbf{2}_{10,10}$ & $\mathbf{2}_{10,10}$ $\rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}})$.

For all except cell 12 the rank vectors are the ones we have seen before.

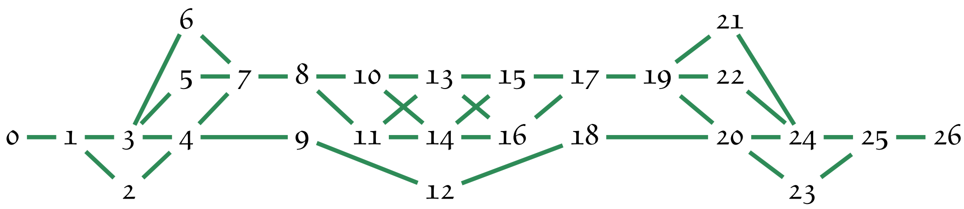
Remark 8.1. Again, in case of the cell 12, there is a discrepancy of numbers similar to the one in Remark 5.1. This comes from the fact that $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$ has rank vector $\mathrm{d}(\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}) = (1,1,2,2,2,4)$, cf. Example 2.9. ▲

8.3. **Type** D_6**.** Graph:



Number of elements: 23040. Number of cells: 27, named 0 (trivial) to 26 (top). Number of parabolic subgroups: 64.

Cell order:



Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
size	1	36	225	25	882	100	100	3650	900	2025	900	100	1600	1600	1952	900	100	900	2025	3650	882	100	100	225	25	36	1
a	0	1	2	2	3	3	3	4	5	6	6	6	7	7	7	8	12	9	10	10	13	15	15	20	20	25	30
p	1	6	9	1	8	1	1	11	2	4	2	1	2	2	2	1	1	1	0	3	1	1	1	0	1	0	1
sr	yes	yes	yes	yes	no	yes	yes	no	yes	yes	yes	yes	yes	yes	no	yes	yes	yes	yes	no	no	yes	yes	yes	yes	yes	yes

Cells 4 and 20 are:

$\mathbf{2}_{9,9}$	$\mathbf{1}_{9,15}$
$\mathbf{1}_{15,9}$	$\mathbf{2}_{15,15}$

$\mathbf{2}_{9,9}$ & $\mathbf{2}_{15,15}$ $\rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}})$,

$\mathbf{2}_{9,9}$	$\mathbf{1}_{9,15}$
$\mathbf{1}_{15,9}$	$\mathbf{2}_{15,15}$

$\mathbf{2}_{9,9}$ & $\mathbf{2}_{15,15}$ $\rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}})$.

Cells 7 and 19 are:

$\mathbf{2}_{5,5}$	$\mathbf{1}_{40,5}$
$\mathbf{1}_{5,40}$	$\mathbf{2}_{40,40}$

$\mathbf{2}_{5,5}$ & $\mathbf{2}_{40,40}$ $\rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}})$,

$\mathbf{2}_{5,5}$	$\mathbf{1}_{40,5}$
$\mathbf{1}_{5,40}$	$\mathbf{2}_{40,40}$

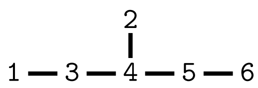
$\mathbf{2}_{5,5}$ & $\mathbf{2}_{40,40}$ $\rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}})$.

Cells 14 is:

$$\begin{array}{|c|c|} \hline \mathbf{2}_{16,16} & \mathbf{1}_{20,16} \\ \hline \mathbf{1}_{16,20} & \mathbf{2}_{20,20} \\ \hline \end{array} \qquad \mathbf{2}_{16,16} \;\&\; \mathbf{2}_{20,20} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Nothing new for the rank vectors in this case.

8.4. **Type** E_6**.** Graph:



Number of elements: 51840. Number of cells: 17, named 0 (trivial) to 16 (top). Number of parabolic subgroups: 64.

Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
size	1	36	400	1350	4096	3600	6561	576	18600	576	6561	3600	4096	1350	400	36	1
a	0	1	2	3	4	5	6	6	7	12	10	11	13	15	20	25	36
p	1	6	10	10	10	5	5	1	5	1	4	2	0	1	2	0	1
sr	yes	yes	yes	no	yes	yes	yes	yes	no	yes	yes	yes	yes	no	yes	yes	yes

Cells 3 and 13 are:

$$\begin{array}{|c|c|} \hline \mathbf{2}_{15,15} & \mathbf{1}_{15,15} \\ \hline \mathbf{1}_{15,15} & \mathbf{2}_{15,15} \\ \hline \end{array} \qquad \mathbf{2}_{15,15} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}),$$

$$\begin{array}{|c|c|} \hline \mathbf{2}_{15,15} & \mathbf{1}_{15,15} \\ \hline \mathbf{1}_{15,15} & \mathbf{2}_{15,15} \\ \hline \end{array} \qquad \mathbf{2}_{15,15} \;\&\; \mathbf{2}_{15,15} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cell 8 is:

$$\begin{array}{|c|c|c|} \hline \mathbf{3}_{10,10} & \mathbf{2}_{50,10} & \mathbf{1}_{20,10} \\ \hline \mathbf{2}_{10,50} & \mathbf{3}_{50,50} & \mathbf{3}_{20,50} \\ \hline \mathbf{1}_{10,20} & \mathbf{3}_{50,20} & \mathbf{6}_{20,20} \\ \hline \end{array} \qquad \begin{array}{l} \mathbf{3}_{10,10} \rightsquigarrow K_0(S_3\text{-}\mathcal{M}\mathrm{od}), \\ \mathbf{3}_{50,50} \rightsquigarrow K_0(S_3\text{-}\mathcal{M}\mathrm{od}), \\ \mathbf{6}_{20,20} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{S_3}). \end{array}$$

Except for the big cell 8 the rank vectors bring no surprise.

Remark 8.2. Since S_3 has subgroups $1, \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}, S_3$ with $\mathbb{Z}/2\mathbb{Z}$ appearing three times, but all are conjugate, and $H^2(S_3, \mathbb{C}^*) = 1$, we get non-equivalent simple transitive 2-representations of $\mathcal{V}\mathrm{ec}_{S_3}$ of ranks $1, 2, 3, 3, 3, 6$. Note however that this pattern does not quite fit to the first and second row of the cell block matrix. The asymptotic Hecke category is not $\mathcal{V}\mathrm{ec}_{S_3}$, but rather S_3-$\mathcal{M}\mathrm{od} \cong \mathrm{SO}(3)_6$ whose ranks of equivalence classes of simple transitive 2-representations read $1, 2, 3, 3$, see Example 2.6. ▲

9. Higher ranks

Up to types E_7 and E_8 there are only three cases.

9.1. **Type** A_n**.** Graph:

$$1 - 2 - 3 - \dots - \mathtt{n}-1 - \mathtt{n}$$

Number of elements: $n!$. Number of cells: $p(n)$ for $p(n)$ being the number of partitions of n. Number of parabolic subgroups: 2^n.

All cells are strongly regular, and the associated asymptotic Hecke algebras and categories are trivial.

9.2. **Type** B_n**.** Graph:

$$1 \overset{4}{-} 2 - 3 - \dots - \mathtt{n}-1 - \mathtt{n}$$

Number of elements: $2^n n!$. Number of cells: $pb(n)$ for $pb(n)$ being the number of certain domino-like partitions of n. Number of parabolic subgroups: 2^n.

For any two left cells L_1, L_2 in a fixed cell J one has $|L_1 \cap L_2^{-1}| = 2^k$ for some $k \in \mathbb{Z}_{>0}$ such that $k^2 + k \leq n$. Moreover, there exists $d(J) \in N$ such that $|L \cap L^{-1}| = 2^d$ for all left cells L in J, and the associated asymptotic Hecke algebra is $K_0(\mathcal{V}\mathrm{ec}_{(\mathbb{Z}/2\mathbb{Z})^d})$, while the associated asymptotic Hecke algebra is $\mathcal{V}\mathrm{ec}_{(\mathbb{Z}/2\mathbb{Z})^d}$.

The rank vectors can be obtained from Example 2.9.

9.3. **Type** D_n**.** Graph:

$$\begin{array}{l} 1 \\ \quad\diagdown \\ \qquad 3 - \dots - \mathtt{n}-1 - \mathtt{n} \\ \quad\diagup \\ 2 \end{array}$$

Number of elements: $2^{n-1} n!$. Number of cells: $pd(n)$ for $pd(n)$ being the number of certain domino-like partitions of n. Number of parabolic subgroups: 2^n.

For any two left cells L_1, L_2 in a fixed cell J one has $|L_1 \cap L_2^{-1}| = 2^k$ for some $k \in \mathbb{Z}_{>0}$ such that $k^2 \leq n$. Moreover, there exists $d(J) \in N$ such that $|L \cap L^{-1}| = 2^d$ for all left cells L in J, and the associated asymptotic Hecke algebra is $K_0(\mathcal{V}\mathrm{ec}_{(\mathbb{Z}/2\mathbb{Z})^d})$, while the associated asymptotic Hecke algebra is $\mathcal{V}\mathrm{ec}_{(\mathbb{Z}/2\mathbb{Z})^d}$.

As for type B_n, the rank vectors can be obtained from Example 2.9.

9.4. **Type** E_7**.** Graph:

$$\begin{array}{ccccccccccccc} & & & & 2 & & & & & & \\ & & & & | & & & & & & \\ 1 & - & 3 & - & 4 & - & 5 & - & 6 & - & 7 \end{array}$$

Number of elements: 2903040. Number of cells: 35, named 0 (trivial) to 34 (top). Number of parabolic subgroups: 128.

Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
size	1	49	729	4802	441	25650	35721	44100	11025	28224	35721	262150	246402	142884	44100	296352	11025	524288
a	0	1	2	3	3	4	5	6	6	6	7	7	8	9	10	10	12	11
p	1	7	15	16	1	20	12	6	1	4	2	12	3	3	1	5	1	5
sr	yes	yes	yes	no	yes	no	yes	yes	yes	yes	yes	no	no	yes	yes	no	yes	no

cell	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
size	11025	296352	44100	142884	246402	262150	35721	28224	11025	44100	35721	25650	441	4802	729	49	1
a	15	13	13	14	15	16	20	21	21	21	22	25	36	30	37	46	63
p	1	1	1	0	2	1	2	1	1	0	0	0	1	1	0	0	1
sr	yes	no	yes	yes	no	no	yes	yes	yes	yes	yes	no	yes	no	yes	yes	yes

Cells 3 and 13 are:

$\mathbf{2}_{21,21}$	$\mathbf{1}_{35,21}$
$\mathbf{1}_{21,35}$	$\mathbf{2}_{35,35}$

$$\mathbf{2}_{21,21}\ \&\ \mathbf{2}_{35,35} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 5 and 29 are:

$\mathbf{2}_{15,15}$	$\mathbf{1}_{105,15}$
$\mathbf{1}_{15,105}$	$\mathbf{2}_{105,105}$

$$\mathbf{2}_{15,15}\ \&\ \mathbf{2}_{105,105} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 11 and 23 are:

$\mathbf{3}_{70,70}$	$\mathbf{2}_{210,70}$	$\mathbf{1}_{35,70}$
$\mathbf{2}_{70,210}$	$\mathbf{3}_{210,210}$	$\mathbf{3}_{35,210}$
$\mathbf{1}_{70,35}$	$\mathbf{3}_{210,35}$	$\mathbf{6}_{35,35}$

$$\mathbf{3}_{70,70} \leftrightsquigarrow K_0(S_3\text{-}\mathcal{M}\mathrm{od}),$$
$$\mathbf{3}_{210,210} \leftrightsquigarrow K_0(S_3\text{-}\mathcal{M}\mathrm{od}),$$
$$\mathbf{6}_{35,35} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{S_3}).$$

Cells 12 and 22 are:

$\mathbf{2}_{189,189}$	$\mathbf{1}_{216,189}$
$\mathbf{1}_{189,216}$	$\mathbf{2}_{216,216}$

$$\mathbf{2}_{189,189}\ \&\ \mathbf{2}_{216,216} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 15 and 19 are:

$\mathbf{2}_{84,84}$	$\mathbf{1}_{336,84}$
$\mathbf{1}_{84,336}$	$\mathbf{2}_{336,336}$

$$\mathbf{2}_{84,84}\ \&\ \mathbf{2}_{336,336} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cell 17 is:

$$\mathbf{2}_{512,512} \qquad \mathbf{2}_{512,512} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

The only interesting rank vectors in this case are the ones for the cells 11 and 23, which can be found in Example 2.5.

9.5. **Type** E_8**.** Graph:

```
            2
            |
1 — 3 — 4 — 5 — 6 — 7 — 8
```

Number of elements: 696729600. Number of cells: 46, named 0 (trivial) to 45 (top). Number of parabolic subgroups: 256.

Size of the cells, Lusztig's a-value (a), number of element of a parabolic subgroup (p) and whether the cells are strongly regular (sr):

cell	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
size	1	64	1225	20384	72200	313600	321489	740000	4986240	5696250	10497600	7768224	7683200	33554432	275625	29635200	12740000	20575296	8037225	36905625	17640000	47360000	4410000
a	0	1	2	3	4	5	6	6	7	8	9	10	10	11	12	12	13	13	14	14	15	15	20
p	1	8	21	28	35	28	7	16	28	12	10	6	4	12	1	6	2	4	1	0	1	4	2
sr	yes	yes	yes	no	no	yes	yes	yes	no	no	no	yes	no	no	no	yes	no	no	yes	yes	yes	yes	yes

cell	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
size	202671840	47360000	17640000	36905625	8037225	20575296	12740000	29635200	275625	33554432	7683200	7768224	10497600	5696250	4986240	740000	321489	313600	72200	20384	1225	64	1
a	16	21	21	22	22	23	25	24	36	26	28	30	31	32	37	42	46	47	52	63	74	91	120
p	4	3	3	0	1	1	0	0	1	0	1	1	0	0	1	1	0	0	0	1	0	0	1
sr	no	yes	yes	yes	yes	no	no	yes	no	no	no	yes	no	no	no	no	yes	yes	no	no	yes	yes	yes

Cells 3 and 42 are:

$\mathbf{2}_{28,28}$	$\mathbf{1}_{84,28}$
$\mathbf{1}_{28,84}$	$\mathbf{2}_{84,84}$

$$\mathbf{2}_{28,28}\ \&\ \mathbf{2}_{84,84} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 4 and 41 are:

$\mathbf{2}_{50,50}$	$\mathbf{1}_{160,50}$
$\mathbf{1}_{50,160}$	$\mathbf{2}_{160,160}$

$$\mathbf{2}_{50,50}\ \&\ \mathbf{2}_{160,160} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 7 and 38 are:

$\mathbf{2}_{300,300}$	$\mathbf{1}_{400,300}$
$\mathbf{1}_{300,400}$	$\mathbf{2}_{400,400}$

$$\mathbf{2}_{300,300}\ \&\ \mathbf{2}_{400,400} \leftrightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 8 and 37 are:

$\mathbf{3}_{448,448}$	$\mathbf{2}_{896,448}$	$\mathbf{1}_{56,448}$
$\mathbf{2}_{448,896}$	$\mathbf{3}_{896,896}$	$\mathbf{3}_{56,896}$
$\mathbf{1}_{448,56}$	$\mathbf{3}_{896,56}$	$\mathbf{6}_{56,56}$

$$\mathbf{3}_{448,448} \rightsquigarrow K_0(S_3\text{-}\mathcal{M}\mathrm{od}),\quad \mathbf{3}_{896,896} \rightsquigarrow K_0(S_3\text{-}\mathcal{M}\mathrm{od}),\quad \mathbf{6}_{56,56} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{S_3}).$$

Cells 9 and 36 are:

$\mathbf{3}_{175,175}$	$\mathbf{2}_{875,175}$	$\mathbf{1}_{350,175}$
$\mathbf{2}_{175,875}$	$\mathbf{3}_{875,875}$	$\mathbf{3}_{350,875}$
$\mathbf{1}_{175,350}$	$\mathbf{3}_{875,350}$	$\mathbf{6}_{350,350}$

$$\mathbf{3}_{175,175} \rightsquigarrow K_0(S_3\text{-}\mathcal{M}\mathrm{od}),\quad \mathbf{3}_{875,875} \rightsquigarrow K_0(S_3\text{-}\mathcal{M}\mathrm{od}),\quad \mathbf{6}_{350,350} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{S_3}).$$

Cells 11 and 34 are:

$\mathbf{2}_{972,972}$	$\mathbf{1}_{1296,972}$
$\mathbf{1}_{972,1296}$	$\mathbf{2}_{1296,1296}$

$$\mathbf{2}_{972,972} \,\&\, \mathbf{2}_{1296,1296} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 12 and 33 are:

$\mathbf{2}_{840,840}$	$\mathbf{1}_{1400,840}$
$\mathbf{1}_{840,1400}$	$\mathbf{2}_{1400,1400}$

$$\mathbf{2}_{840,840} \,\&\, \mathbf{2}_{1400,1400} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 11 and 26 are :

$$\mathbf{2}_{4096,4096} \qquad \mathbf{2}_{4096,4096} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 15 and 30 are:

$\mathbf{2}_{840,840}$	$\mathbf{1}_{3360,840}$
$\mathbf{1}_{840,3360}$	$\mathbf{2}_{3360,3360}$

$$\mathbf{2}_{840,840} \,\&\, \mathbf{2}_{3360,3360} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 16 and 29 are:

$\mathbf{2}_{700,700}$	$\mathbf{1}_{2100,700}$
$\mathbf{1}_{700,2100}$	$\mathbf{2}_{2100,2100}$

$$\mathbf{2}_{700,700} \,\&\, \mathbf{2}_{2100,2100} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cells 21 and 25 are:

$\mathbf{2}_{2400,2400}$	$\mathbf{1}_{3200,2400}$
$\mathbf{1}_{2400,3200}$	$\mathbf{2}_{3200,3200}$

$$\mathbf{2}_{2400,2400} \,\&\, \mathbf{2}_{3200,3200} \rightsquigarrow K_0(\mathcal{V}\mathrm{ec}_{\mathbb{Z}/2\mathbb{Z}}).$$

Cell 23 is:

To be done, but the asymptotic category is S_5-$\mathcal{M}\mathrm{od}$

The rank vectors for the cases of S_3-$\mathcal{M}\mathrm{od}$ and S_5-$\mathcal{M}\mathrm{od}$ can be found in Example 2.5.

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